

The University of Nottingham Malaysia Campus

SCHOOL OF COMPUTER SCIENCE

A LEVEL 3 MODULE, AUTUMN SEMESTER 2020-2021

Machine Learning (COMP3038)

Time allowed TWO Hours

Candidates may complete the front cover of their answer book and sign their desk card but must NOT write anything else until the start of the examination period is announced

Answer All 3 Questions

Dictionaries are not allowed with one exception. Those whose first language is not English may use a standard translation dictionary to translate between that language and English provided that neither language is the subject of this examination. Subject specific translation dictionaries are not permitted.

No electronic devices capable of storing and retrieving text, including electronic dictionaries, may be used.

DO NOT turn your examination paper over until instructed to do so

ADDITIONAL MATERIAL:

NONE

INFORMATION FOR INVIGILATORS:

NONE

Question 1: Foundations of Machine Learning [overall 34 marks]

- a) Give the formal definition of cost function and name a method that can minimize the cost function. In certain cost minimisation methods there's a learning rate, which is a tuning parameter in the first part of the optimization of cost function, explain what potential problems would be when setting a learning rate.

[6 marks]

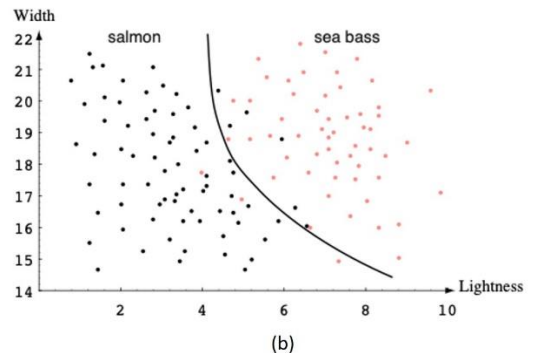
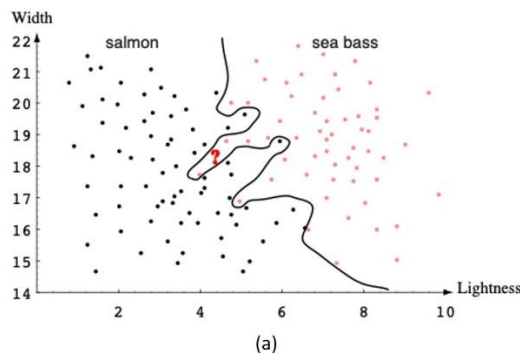
- b) In linear regression, use set and numerical domain notation to denote basis functions and explain why we need such functions.

[4 marks]

- c) (i) Explain the difference between a linear classifier and a non-linear classifier. (ii) Draw two diagrams: one with a set of 2-dimensional data points of two classes for which a linear classifier would be a good solution, and one where a non-linear classifier would be the better choice. (iii) For the non-linear classifier in input space example, draw a diagram to make the linear classification in feature space by applying a kernel mapping method.

[11 marks]

- d) Which decision boundary would you prefer in figure (A) and justify your answer.



(A)**[6 marks]**

- e) Explain what the Gold Price Prediction problem is in the context of machine learning and what T, P & E are?

[7 marks]

Question 2: Unsupervised Learning [overall 33 marks]

Table 1 Data Sample 1

Instance	Attribute 1	Attribute 2
3	3	4
2	3	6
3	3	8
4	4	5
5	4	7
6	5	1
7	5	5
8	7	3
9	7	5
10	8	5

Table 2 Cluster Center

Cluster centre	Attribute 1	Attribute 2
mu1	1	3
mu2	9	4

Table 3 Data Sample 2

Instance	Attribute 1	Attribute 2	Attribute 3
1	Y	N	Y
2	N	Y	N
3	Y	N	N

- a) Consider a Gaussian mixture model in which the marginal distribution $p(z)$ for the latent variable is given by (2.1), and the conditional distribution $p(x|z)$ for the observed variable is given by (2.2). Show that the marginal distribution $p(x)$, obtained by summing $p(z)p(x|z)$ over all possible values of z , is a Gaussian mixture of the form (2.0).

$$p(x) = \sum_{k=1}^K \pi_k N(\mathbf{x}|\boldsymbol{\mu}_k, \Sigma_k) \quad (2.0)$$

$$p(\mathbf{z}) = \prod_{k=1}^K \pi_k^{z_k} \quad (2.1)$$

$$p(\mathbf{x}|\mathbf{z}) = \prod_{k=1}^K N(\mathbf{x}|\boldsymbol{\mu}_k, \Sigma_k)^{z_k} \quad (2.2)$$

[6 marks]

- b) Draw diagrams to show how K-Means works for data sample 1 in **Table 1** with the initial cluster centers in **Table 2**. Identify 3 problems with such K-Means implementation and give your solutions.

[10 marks]

- c) If one or both of the attributes in **Table 1** represent categorical labels, can K-Means still be implemented for this clustering task? Justify your answer and suggest another clustering algorithm for this case.

[4 marks]

- d) Let's suppose we are solving a clustering problem over data sample 2 in **Table 3** with 3 binary attribute 1, attribute 2 and attribute 3. How many distinct instances we can get? What if we add 7 more binary attributes attribute 4, attribute 5, ..., attribute 10, how many distinct instances can we get? Elaborate what the potential problem is when adding more attributes to the data sample and propose your solution (with details) to solve this problem.

[6 marks]

- e) If we want to implement EM to carry out the clustering task works for data sample 1 in **Table 1**, what are the tasks of the E-step and M-step of the EM-algorithm? And explain when the EM iterations will stop?

[7 marks]

Question 3: Evaluating Hypotheses and Decision Tree [overall 34 marks]

- a) We trained two classifiers, Decision Tree with 350 examples and tested with 30 examples; and the Support Vector Machine with 500 examples and tested with 50 examples. The correctly classified percentage of the Decision Tree is 89% and for the support vector machine it is 92%. Calculate the confidence intervals of the two true errors and comment how that relates to the statistical difference between the results.

$Z_n = 1.96$ for a confidence level of 95%.

[6 marks]

- b) Use tossing a coin as an example to justify how the expected difference between sample error and true error depends on the size of the data sample.

[10 marks]

- c) Suppose we implement a Decision Tree algorithm with 10-fold Cross Validation for a classification problem and the confusion matrix is shown in **Table 4**. State the total number of instances, correctly classified instances, and what the recall and precision rates for class C are.

Table 4 Confusion Matrix

Predicted True	A	B	C
A	50	30	10
B	20	30	10
C	0	10	50

[4 marks]

- d) Considering the example in **Table 5**, what is the maximal number of leaf (decision) nodes that we can have in a decision tree for this data? What is the maximal possible depth of a decision tree for this data? Draw the tree, explain which attribute should be the root node and why?

Table 5 Training Set Example

Instance	Classification	Attribute 1	Attribute 2
1	C1	Y	Y
2	C1	Y	Y
3	C2	Y	N
4	C1	N	N
5	C2	N	Y
6	C2	N	Y

[10 marks]

- e) Using the example in **Table 5** to prove that the decrease in entropy impurity provided by a single yes/no query can never be greater than one bit.

[4 marks]